



BUS-1023 Applied Mathematics

CLO 2.3

Compute Gross Pay on the basis of time.

Monday, September 21, 2026



Terms of salary payment	Period (based on a year)
Weekly	52 times (once a week)
Biweekly	26 times (every two weeks)
Semimonthly	24 times (twice a month)
Monthly	12 times (once a month)

Let's assume you earn **\$50,000 per year**. The following table shows what you would earn each pay period.

Terms of salary payment	Period (based on a year)	Earnings for period (dollars)
Weekly	52 times (once a week)	\$ 961.54 ($\$50,000 \div 52$)
Biweekly	26 times (every two weeks)	\$1,923.08 ($\$50,000 \div 26$)
Semimonthly	24 times (twice a month)	\$2,083.33 ($\$50,000 \div 24$)
Monthly	12 times (once a month)	\$4,166.67 ($\$50,000 \div 12$)



Example 1a: Mariam has an annual salary of AED 362,700. If she is paid on a weekly basis. How much is her weekly rate?

Solution:

$$\text{Salary rate} = \frac{\text{annual salary}}{\text{number of periods in a year}} \quad \text{Mariam's Weekly rate} = \frac{\text{AED } 362,700}{52} = \text{AED } 6975$$

Example 1b: Ahmed receives an amount of AED 7,400 monthly for his housing allowance. If he plans to take his annual allowance in advance, how much will he be able to receive?

Solution:

Annual amount = *Salary rate x number of periods in a year*

$$\text{Ahmed's annual allowance} = \text{AED } 7400 \times 12 = \text{AED } 88,800$$





Hourly Pay



Employees – People who work for others

Employer – The person or company and employee works for.

Hourly Rate - a certain amount of money a person is paid per hour they work.

Example: A worker at McDonald's makes **\$7.35 an** hour for every hour they work

Gross Pay

Gross Pay – the total amount of money that an employee is paid.

This is also called **total earnings**.





What is meant by **Overtime**? (Getting extra pay for extra work)

There are two methods to calculate overtime:

1. Any hours worked more than 40 hours a week.
2. Any hours over a stated amount of hours per day.

Most people are paid **time and half (1.5)** for every overtime hour worked.

To calculate overtime pay:

Overtime Pay = Pay per hour x 1.5 x number of overtime hours.





To calculate Gross Pay with Overtime Hours:

If the employee works **more than 40 hours**:

$$\text{Regular Pay} = 40 \times \$ \text{ rate per hour}$$

$$\text{Over Time Hours} = \text{No. of hours worked} - 40$$

$$\text{Over Time Pay} = \$ \text{ rate per hour} \times 1.5 \times \text{Over Time Hours}$$

$$\text{Gross Pay} = \text{Regular Pay} + \text{Over Time pay}$$





Example 1:

Tom worked 45 hours and earns \$10 an hour. How much is his **gross pay** for the week?

$$\text{Regular Pay} = 40 \times \$10 = \underline{\$ 400}$$

$$\text{Over Time Hours} = 45 - 40 = \underline{5 \text{ hours}}$$

$$\text{Over Time Pay} = \$10 \times 1.5 \times 5 = \underline{\$ 75}$$

$$\text{Gross Pay} = \$400 + \$75 = \underline{\$ 475}$$





Example 2:

Joel worked 52 hours and earns \$20 an hour. How much is his **gross pay** for the week?

$$\text{Regular Pay} = 40 \times \$20 = \underline{\$ 800}$$

$$\text{Over Time Hours} = 52 - 40 = \underline{12 \text{ hours}}$$

$$\text{Over Time Pay} = \$20 \times 1.5 \times 5 = \underline{\$ 360}$$

$$\text{Gross Pay} = \$800 + \$360 = \underline{\$ 1,160}$$





Calculation of Overtime

Gross pay = Regular Pay + Overtime Pay

Regular pay (40-hour week) = *No. of hours × Regular rate per hour*

Hourly overtime pay rate = *Regular hourly pay rate × 1.5*

Overtime pay = *Overtime Pay Rate × Overtime Hours*

Gross pay = *Regular Pay + Overtime Pay*





Calculation of Overtime

Gross pay = Regular Pay + Overtime Pay

Example 3:

Calculate Ramon Valdez's Gross Pay from the following table of data.

His regular pay rate is \$9 per hour.

Assume he works a standard 40 hour week.

Employee	M	T	W	Th	F	S	Total
R Valdez	13	8.5	10	8	11.25	10.75	61.5

$$\begin{array}{r} 61\frac{1}{2} \text{ total hours} \\ -40 \text{ regular hours} \\ \hline 21\frac{1}{2} \text{ hours overtime}^1 \end{array} \quad \text{Time-and-a-half pay: } \$9 \times 1.5 = \$13.50$$

$$\begin{aligned} \text{Gross pay} &= (40 \text{ hours} \times \$9) + (21\frac{1}{2} \text{ hours} \times \$13.50) \\ &= \$360 + \$290.25 \\ &= \$650.25 \end{aligned}$$





Practice:

11

- 1) Abdullah worked 52 hours in one week for a local company. If he receives AED 100 per hour. What is Abdullah's gross pay, assuming overtime is paid at time-and-half?
Overtime begins after 40 working hours.

Answer: **AED 5,800**





Sara earns AED 32 per hour and worked 48 hours this week. Overtime is paid at 1.5 times the regular hourly rate.

Calculate:

- a. Regular pay
- b. Overtime pay
- c. Total gross pay





Omar earns AED 38 per hour and worked 50 hours this week. His company pays 1.5 times the regular rate for all hours beyond 40.

Determine:

- a. His overtime hourly rate
- b. His overtime pay
- c. His total gross pay



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